

Sanjeev Kumar

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LinkedIn | GitHub | LeetCode | Portfolio

EDUCATION

IILM University, Greater Noida
B.Tech in Computer Science and Engineering

2023 - 2027
CGPA: 7.89/10

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL
Frontend: Next.js, React.js, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
Databases: MongoDB, PostgreSQL, Prisma
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
Tools & Platforms: Git, GitHub, AWS, Postman, VS Code, Chrome Extension Development

PROJECTS

INORA - Multi-Channel EEG-Based Control **Dec 2025 - Jan 2026**
Technologies: AD620, TL072, ADS1115, ESP32, Python, NumPy

- Built a low-cost **3-channel EEG acquisition system** with wireless streaming at **83 Hz/channel**, and developed a Python pipeline for EEG acquisition, filtering, and DRL-based noise reduction, validated on **20 volunteers**.
- Demonstrated EEG-based Pong control with valid **15–80 uV signals** using hardware costing **INR 3,146**.

BrainScore - EEG-Based Depression Detection **Mar 2025 - Nov 2025**
Technologies: Python, Pandas, Scikit-learn, LightGBM, SHAP, Google Colab

- Processed EEG recordings from **58 subjects** using **20-channel signals** and extracted **18 statistical, frequency, and entropy-based biomarkers**; built and evaluated **6 machine learning models**.
- Achieved **95.28% classification accuracy** using **LightGBM**, with **SHAP** analysis to interpret feature importance and improve model explainability for automated depression detection.

MUNC Dashboard - Kasper Infotech User Dashboard **June 2025 - Aug 2025**
Technologies: JavaScript, Next.js, React.js

- Developed and enhanced responsive dashboard pages using **JavaScript, Next.js, and React.js**, implementing frontend features and improving UI usability across multiple pages.
- Resolved application bugs and supported testing and documentation to improve reliability and maintainability.

EXPERIENCE

Frontend React Developer - Intern **Dec 2025 - Feb 2026**
Kasper Infotech Pvt. Ltd. *Remote*

- Developed and improved responsive web pages using **React.js**, built reusable UI components, and enhanced interface structure, usability, and responsiveness.
- Resolved UI and frontend bugs while maintaining technical documentation and project updates.

Machine Learning & Data Science Intern **May 2025 - Jul 2025**
Svaarogyam Medical Devices Pvt. Ltd. *Remote*

- Worked on healthcare machine learning tasks involving **data cleaning, preprocessing, feature extraction**, and predictive modeling for medical device applications.
- Built end-to-end **ML pipelines** collaboratively with a remote team to deliver project milestones in the medical domain.

ACHIEVEMENTS / CERTIFICATIONS

- Published Patent Application, Indian Patent Office (Application No. 202611071187): *Wearable Head-Mounted Device for Continuous Mental State Monitoring and Adaptive Cognitive Regulation* **10 Aug 2026**
- Research article accepted in **IEEE Transactions on Medical Robotics and Bionics (T-MRB)**: *SHAP-based Study: XAI-Guided Optimisation of Spatial-Temporal Bio-Neuro Marker EEG for Depression* **19 Jul 2026**
- Research Paper accepted at **IEEE Sponsored ETNCC 2026**: *INORA: A Low-Cost 3-Channel EEG Acquisition System with Driven-Right-Leg Noise Suppression for BCI Prototyping* **31 May 2026**
- Hackathon Winner (1st Rank)** among 1,004 participants across 118 teams **02 Apr 2026**
- Secured **1st Place in the Call for Paper event** at SRM Institute of Science and Technology **17 Sep 2025**